

Functional Requirements Document

ReqAgent — AI Requirement Intake & Tracker

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1. Purpose

This document defines the functional requirements for the ReqAgent system. It translates business requirements from the BRD into specific, testable system behaviors.

2. System Modules

Module	Description
Intake Form	Stakeholder-facing form for requirement submission
AI Processor	LLM-powered analysis and story generation (Groq LLaMA 3.3 70B)
Data Layer	Dual-database storage (MongoDB Atlas + Neon Postgres)
Dashboard	Four-tab interface for tracking, traceability, and analytics

3. Functional Requirements

3.1 Intake Form Module

ID	Requirement	BR Ref
FR-01	System shall provide text input for requirement title (max 500 chars)	BR-01
FR-02	System shall provide free-text area for requirement description	BR-01
FR-03	System shall provide dropdown for department selection	BR-01
FR-04	System shall provide dropdown for submitter role selection	BR-01
FR-05	System shall provide text input for business objective	BR-01

ID	Requirement	BR Ref
FR-06	System shall provide optional text area for additional notes	BR-01
FR-07	System shall validate all required fields before submission	BR-01
FR-08	System shall display AI results immediately after submission	BR-02

3.2 AI Processing Module

ID	Requirement	BR Ref
FR-09	System shall generate exactly 3 user stories per requirement	BR-02
FR-10	Stories shall follow "As a [user], I want [action] so that [benefit]" format	BR-02
FR-11	Each story shall include 2-3 acceptance criteria in Given/When/Then format	BR-02
FR-12	System shall assign one MoSCoW classification per requirement	BR-03
FR-13	System shall assign priority (High/Medium/Low) with written justification	BR-04
FR-14	System shall scan all existing requirements for conflicts on each submission	BR-05
FR-15	System shall return conflict description or "No conflicts detected"	BR-05
FR-16	AI processing shall complete within 10 seconds	BR-02

3.3 Data Layer Module

ID	Requirement	BR Ref
FR-17	System shall store raw intake document in MongoDB Atlas	BR-10
FR-18	MongoDB document shall include title, description, department, role, objective, notes, timestamp	BR-10
FR-19	System shall store structured user stories in Neon Postgres	BR-10
FR-20	Postgres shall maintain requirements table with status, priority, MoSCoW columns	BR-10
FR-21	Postgres shall maintain user_stories table linked to requirements	BR-10
FR-22	Postgres shall maintain status_audit table for all status changes	BR-07
FR-23	Postgres shall maintain conflicts table for detected conflicts	BR-05
FR-24	MongoDB document ID shall be stored in Postgres for cross-reference	BR-10

3.4 Requirements Tracker Module

ID	Requirement	BR Ref
FR-25	System shall display all requirements in an expandable list	BR-06
FR-26	System shall support filtering by status, priority, and MoSCoW	BR-06
FR-27	System shall display status, priority, MoSCoW, submitter, department per requirement	BR-06
FR-28	System shall allow status updates via dropdown	BR-06
FR-29	Status options: Submitted, Under Review, Approved, In Development, Done, Rejected	BR-06
FR-30	System shall log every status change with old status, new status, and timestamp	BR-07
FR-31	System shall display full user stories and acceptance criteria on demand	BR-02
FR-32	System shall display original raw intake from MongoDB on demand	BR-10
FR-33	System shall display complete audit trail per requirement	BR-07

3.5 Traceability Matrix Module

ID	Requirement	BR Ref
FR-34	System shall display a matrix linking requirements to user stories	BR-08
FR-35	Matrix shall include REQ ID, title, submitter, department, status, priority, MoSCoW, story, criteria	BR-08
FR-36	Matrix shall support filtering by status and MoSCoW	BR-08
FR-37	Matrix shall apply color coding to status, priority, and MoSCoW columns	BR-08
FR-38	System shall provide CSV export of the full matrix	BR-08

3.6 Analytics Module

ID	Requirement	BR Ref
FR-39	System shall display KPI metrics: total, approved, high priority, must have	BR-09
FR-40	System shall display requirements by status as donut chart	BR-09
FR-41	System shall display MoSCoW breakdown as pie chart	BR-09
FR-42	System shall display requirements by department as horizontal bar chart	BR-09
FR-43	System shall display requirements by priority as bar chart	BR-09
FR-44	System shall display submissions over time as line chart	BR-09
FR-45	System shall display submissions by role as bar chart	BR-09

4. Non-Functional Requirements

ID	Requirement	Category
NFR-01	App shall load within 3 seconds on standard connection	Performance
NFR-02	AI processing shall complete within 10 seconds	Performance
NFR-03	API keys shall never be stored in source code	Security
NFR-04	All database connections shall use SSL	Security
NFR-05	App shall be deployable on Streamlit Cloud	Portability
NFR-06	Codebase shall follow modular architecture	Maintainability

5. Data Dictionary

MongoDB — raw_intake collection

Field	Type	Description
_id	ObjectId	Auto-generated document ID
title	String	Requirement title
description	String	Full requirement description
department	String	Submitting department
submitted_by	String	Submitter name
role	String	Submitter role
business_objective	String	Business objective
additional_notes	String	Optional notes
submitted_at	DateTime	Submission timestamp
processed	Boolean	Whether AI has processed this document

Neon Postgres — requirements table

Column	Type	Description
id	SERIAL PK	Auto-incremented requirement ID
mongo_id	VARCHAR	Reference to MongoDB document
title	VARCHAR	Requirement title

Column	Type	Description
submitted_by	VARCHAR	Submitter name
department	VARCHAR	Department
role	VARCHAR	Submitter role
status	VARCHAR	Current lifecycle status
priority	VARCHAR	AI-assigned priority
moscow	VARCHAR	MoSCoW classification
created_at	TIMESTAMP	Creation timestamp
updated_at	TIMESTAMP	Last update timestamp